

AOS-CX 10.06 IP Services Guide

6100 Switch Series



Hewlett Packard
Enterprise

Part Number: 5200-7777
Published: January 2021
Edition: 1

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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in **Support and other resources**.

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: <code><example-text></code> <i>example-text</i> <code>example-text</code> <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: - For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. - For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.

Table Continued

Convention	Usage
[]	Brackets. Indicates that the enclosed item or items are optional.
... or	Ellipsis:
...	- In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. - In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment. The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the interface context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.

ICMP Router Discovery Protocol (IRDP), an extension of the ICMP, is independent of any routing protocol. It allows hosts to discover the IP addresses of neighboring routers that can act as default gateways to reach devices on other IP networks.



NOTE: On the switches covered by this guide, IRDP is configured on a VLAN interface.

IRDP operation

IRDP uses the following types of ICMP messages:

- Router advertisement (RA): Sent by a router to advertise IP addresses (including the primary and secondary IP addresses) and preference.
- Router solicitation (RS): Sent by a host to request the IP addresses of routers on the subnet.

An interface with IRDP enabled periodically broadcasts or multicasts an RA message to advertise its IP addresses. A receiving host adds the IP addresses to its routing table, and selects the IP address with the highest preference as the default gateway.

When a host attached to the subnet starts up, the host multicasts an RS message to request immediate advertisements. If the host does not receive any advertisements, it retransmits the RS several times. If the host does not discover the IP addresses of neighboring routers because of network problems, the host can still discover them from periodic RAs.

IRDP allows hosts to discover neighboring routers, but it does not suggest the best route to a destination. If a host sends a packet to a router that is not the best next hop, the host will receive an ICMP redirect message from the router.

IP address preference

Every IP address advertised in RAs has a preference value. A larger preference value represents a higher preference. The IP address with the highest preference is selected as the default gateway address.

You can specify the preference for IP addresses to be advertised on a router interface.

An address with the minimum preference value (-2147483648) will not be used as a default gateway address.

Lifetime of an IP address

An RA contains a lifetime field that specifies the lifetime of advertised IP addresses. If the host does not receive a new RA for an IP address within the address lifetime, the host removes the route entry.

All the IP addresses advertised by an interface have the same lifetime.

Advertising interval

A router interface with IRDP enabled sends out RAs randomly between the minimum and maximum advertising intervals. This mechanism prevents the local link from being overloaded by a large number of RAs sent simultaneously from routers.

As a best practice, shorten the advertising interval on a link that suffers high packet loss rates

Destination address of RA

An RA uses either of the following destination IP addresses:

- Broadcast address 255.255.255.255.
- Multicast address 224.0.0.1, which identifies all hosts on the local link.

By default, the destination IP address of an RA is the multicast address. If all listening hosts in a local area network support IP multicast, specify 224.0.0.1 as the destination IP address.

Proxy-advertised IP addresses

By default, an interface advertises its primary and secondary IP addresses. You can specify IP addresses of other gateways for an interface to proxy-advertise.

VRF support

In IP-based computer networks, virtual routing and forwarding (VRF) is a technology that allows multiple instances of a routing table to co-exist within the same router at the same time. Because the routing instances are independent, the same or overlapping IP addresses can be used without conflicting with each other.

IRDP is VRF aware. As the router advertisements and solicit processing occurs on the interface, packet is through the interface and corresponding VRF.

VSX synchronization

IRDP supports VSX synchronization. For more information on using VSX, see the *Virtual Switching Extension (VSX) Guide* for your switch and software version

Configuring IRDP

Prerequisites

A layer 3 interface.

Procedure

1. Enable IRDP on an interface with the command .
2. Set the maximum hold time with the command .
3. Set the maximum router advertisement interval with the command .
4. Set the minimum router advertisement interval with the command .
5. Set the IRDP preference level with the command .
6. Review IRDP configuration settings with the command .

Example

This example creates the following configuration:

- Enables IRDP on the layer 3 VLAN interface 2 with packet type set to broadcast.
- Sets the hold time to 5000 seconds.
- Sets the advertisement interval to 30 seconds.

- Sets the minimum advertisement interval to 25 seconds.
- Sets the IRDP preference level to 25.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp broadcast
switch(config-if-vlan)# ip irdp holdtime 5000
switch(config-if-vlan)# ip irdp maxadvertinterval 30
switch(config-if-vlan)# ip irdp minadvertinterval 25
switch(config-if-vlan)# ip irdp preference 25
```

IRDP commands

diag-dump irdp basic

Syntax

```
diag-dump irdp basic
```

Description

Displays diagnostic information for IRDP.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# diag-dump irdp basic
=====
[Start] Feature irdp Time : Thu Jan  7 04:46:25 2021
=====
-----
[Start] Daemon hpe-rdiscd
-----
Interface: vlan2 (state : Down)
rdisc ipv4 (enabled: 1, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
Interface: vlan1 (state : Down)
rdisc ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
-----
[End] Daemon hpe-rdiscd
-----
=====
[End] Feature irdp
=====
Diagnostic-dump captured for feature irdp
```

ip irdp

Syntax

```
ip irdp [broadcast | multicast]
```

```
no ip irdp
```

Description

Enables IRDP on an interface and specifies the packet type that is used to send advertisements. By default, the packet type is set to `multicast`. IRDP is only supported on layer 3 interfaces.

The `no` form of this command disables IRDP on an interface.

Command context

```
config-if
```

Parameters

broadcast

Advertisements are sent as broadcast packets to IP address 255.255.255.255.

multicast

Advertisements are sent as multicast packets to the multicast group with IP address 24.0.0.1. Default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling IRDP on interface vlan 2 with packet type set to the default value (multicast).

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp
```

Enabling IRDP on interface vlan 2 with packet type set to broadcast.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp broadcast
```

Disabling IRDP.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp
```

ip irdp holdtime

Syntax

```
ip irdp holdtime <TIME>
```

Description

Specifies the maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives. When a new advertisement arrives, hold time is reset. Hold time must be greater than or equal to the maximum advertisement interval. Therefore, if the hold time for an advertisement expires, the host can reasonably conclude that the router interface that sent the advertisement is no longer available. The default hold time is three times the maximum advertisement interval.

Command context

config-if

Parameters

<TIME>

Specifies the lifetime of router advertisements sent from this interface. Range: 4 to 9000 seconds. Default: 1800 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the hold time for VLAN interface 2 to 5000 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp holdtime 5000
```

ip irdp maxadvertinterval

Syntax

ip irdp maxadvertinterval <TIME>

Description

Specifies the maximum router advertisement interval.

Command context

config-if

Parameters

<TIME>

Specifies the maximum time allowed between the sending of unsolicited router advertisements. Range: 4 to 1800 seconds. Default: 600 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the advertisement interval for VLAN interface 2 to 30 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp maxadvertinterval 30
```

ip irdp minadvertinterval

Syntax

ip irdp minadvertinterval <TIME>

Description

Specifies the minimum amount of time the switch waits between sending router advertisements. By default, this value is automatically set by the switch to be 75% of the value configured for maximum router advertisement interval. Use this command to override the automatically configured value.

Command context

config-if

Parameters

<TIME>

Specifies the minimum time allowed between the sending of unsolicited router advertisements. Range: 3 to 1800 seconds. Default: 450 seconds (75% of the default value for maximum router advertisement interval).

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the minimum advertisement interval for VLAN interface 2 to 25 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp minadvertinterval 25
```

ip irdp preference

Syntax

ip irdp preference <LEVEL>

Description

Specifies the IRDP preference level. If a host receives multiple router advertisement messages from different routers, the host selects the router that sent the message with the highest preference as the default gateway.

Command context

config-if

Parameters

<LEVEL>

Specifies the IRDP preference level. Range: -2147483648 to 2147483647. Default: 0.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the IRDP preference level for VLAN interface 2 to 25.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp preference 25
```

show ip irdp

Syntax

```
show ip irdp [vsx-peer]
```

Description

Displays IRDP configuration settings.

Command context

Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# sh ip irdp
```

ICMP Router Discovery Protocol

Interface	Status	Advertising Address	Minimum Interval	Maximum Interval	Holdtime	Preference
-----	-----	-----	-----	-----	-----	-----
vlan1	Disabled	multicast	450	600	1800	0
bridge_normal	Disabled	multicast	450	600	1800	0

IPv6 RA provides a method for local IPv6 hosts to automatically configure their own IP address (and other settings such as a preferred DNS server) based on information advertised by switches/routers operating on the network.

IPv6 flags

Behavior of IPv6 hosts to IPv6 RA messages is controlled by the managed address configuration flag (M flag), and other stateful configuration flag (O flag).

M flag	O flag	Description
0	0	Indicates that no information is available via DHCPv6.
0	1	Indicates that other configuration information is available via DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.
1	0	Indicates that addresses are available via Dynamic Host Configuration Protocol (DHCPv6).
1	1	If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

Configuring IPv6 RA

Procedure

1. Enable transmission of IPv6 router advertisements with the command `no ipv6 nd suppress-ra`.
2. Optionally, configure IPv6 unicast address prefixes with the command `ipv6 nd prefix`.
3. Optionally, configure support for DNS name resolution with the commands `ipv6 nd ra dns server` and `ipv6 nd ra dns search-list`.
4. For most deployments, the default values for the following features do not need to be changed. If your deployment requires different settings, change the default values with the indicated command:

IPv6 RA setting	Default value	Command to change it
Number of neighbor solicitations to be sent when performing DAD.	1	<code>ipv6 nd dad attempts</code>
Number of neighbor entries in the ND cache.	131072	<code>ipv6 nd cache-limit</code>
Hop limit to be sent in the RA messages.	64	<code>ipv6 nd hop-limit</code>
MTU value to be sent in the RA messages.	1500 bytes	<code>ipv6 nd mtu</code>
Neighbor solicitation interval	1000 milliseconds	<code>ipv6 nd ns-interval</code>

Table Continued

IPv6 RA setting	Default value	Command to change it
Lifetime of a default router.	1800 seconds	<code>ipv6 nd ra lifetime</code>
Retrieval of an IPv6 address by devices.	Disabled	<code>ipv6 nd ra managed-config-flag</code>
Maximum interval between transmissions of IPv6 RAs.	600 seconds	<code>ipv6 nd ra max-interval</code>
Minimum interval between transmissions of IPv6 RAs.	200 seconds	<code>ipv6 nd ra min-interval</code>
Time that an interface considers a device to be reachable.	0 milliseconds (no limit)	<code>ipv6 nd ra reachable-time</code>
Retry period between ND solicitations.	0 (Use locally configured NS-interval)	<code>ipv6 nd ra retrans-timer</code>
Default routing preference for an interface.	Medium	<code>ipv6 nd router-preference</code>

5. Review IPv6 RA configuration settings with the commands `show ipv6 nd interface`, `show ipv6 nd interface prefix`, `show ipv6 nd ra dns server`, and `show ipv6 nd ra dns search-list`.

Example

This example creates the following configuration:

- Enables IPV6 RA on interface **interface vlan 2**.
- Sets the recursive DNS server address to **4001::1** with a lifetime of **400** seconds.
- Sets the minimum interval between transmissions to **3** seconds.
- Sets the maximum interval between transmissions to **13** seconds.
- Sets the lifetime of a default router to **1900** seconds.

```
switch(config)# interface vlan 2
switch(config-if)# no ipv6 nd suppress-ra
switch(config-if)# ipv6 nd ra dns server 4001::1 lifetime 400
switch(config-if)# ipv6 nd ra min-interval 3
switch(config-if)# ipv6 nd ra max-interval 13
switch(config-if)# ipv6 nd ra lifetime 1900
switch(config-if)# end
switch# show ipv6 nd interface vlan 2
Interface vlan2 is up
Admin state is up
IPv6 address:
  2006::1/64 [VALID]
IPv6 link-local address: fe80::98f2:b321:368:6dc6/64 [VALID]
ICMPv6 active timers:
  Last Router-Advertisement sent: 0 Secs
  Next Router-Advertisement sent in: 13 Secs
Router-Advertisement parameters:
  Periodic interval: 3 to 13 secs
  Router Preference: medium
  Send "Managed Address Configuration" flag: false
  Send "Other Stateful Configuration" flag: false
```

```

    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1900
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: false
    Suppress MTU in RA: true
    ICMPv6 error message parameters:
    Send redirects: false
    ICMPv6 DAD parameters:
    Current DAD attempt: 1
switch# show ipv6 nd ra dns server
Recursive DNS Server List on: vlan2
    Suppress DNS Server List: No
    DNS Server 1: 2001::1    lifetime 400

```

IPv6 RA scenario

In this scenario, two host computers are auto-configured with IP addresses using IPv6 RA. In addition, the switch provides the hosts with an address of a recursive DNS server.

Procedure

1. Configure the VLAN interfaces with IPv6 addresses.

```

switch# config
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address 2001::1/64
switch(config)# interface vlan 3
switch(config-if-vlan)# ipv6 address 3001::1/64
switch(config)# interface vlan 4
switch(config-if-vlan)# ipv6 address 4001::1/64

```

2. Enable transmission of all IPv6 RA messages.

```

switch(config-if-vlan)# no ipv6 nd suppress-ra

```

IPv6 RA commands

ipv6 address <global-unicast-address>

Syntax

```

ipv6 address <global-unicast-address>
no ipv6 address <global-unicast-address>

```

Description

Sets a global unicast address on the interface.

The **no** form of this command removes the global unicast address on the interface.



NOTE: This command automatically creates an IPv6 link-local address on the interface. However, it does not add the command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the command.

Command context

config-if

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Enabling a global unicast address:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address 3731:54:65fe:2::a7
```

Disabling a global unicast address:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ipv6 address 3731:54:65fe:2::a7
```

ipv6 address autoconfig

Syntax

```
ipv6 address autoconfig
no ipv6 address autoconfig
```

Description

Enables the interface to automatically obtain an IPv6 address using router advertisement information and the EUI-64 identifier.

The `no` form of this command disables address auto-configuration.



NOTE:

- A maximum of 15 autoconfigured addresses are supported.
 - This command automatically creates an IPv6 link-local address on the interface. However, it does not add the command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the command.
-

Command context

config-if

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

The IPv6 SLAAC feature lets the router obtain the IPv6 address for the interface it is configured through the SLAAC method. This feature is not available on the `mgmt` VRF.

Example

Enabling unicast autoconfiguring:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address autoconfig
```

Disabling unicast autoconfiguring:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ipv6 address autoconfig
```

ipv6 address link-local

Syntax

```
ipv6 address link-local [<IPV6-ADDR>/<MASK>]
```

Description

Enables IPv6 on the current interface. If no address is specified, an IPv6 link-local address is auto-generated for the interface. If an address is specified, auto-configuration is disabled and the specified address/mask is assigned to the interface.

To disable IPv6 link-local on the interface, remove `ipv6 address link-local`, `ipv6 address <global-ipv6-address>`, and `ipv6 address autoconfig` from the interface.



NOTE: This feature is not available on the management VRF.

Command context

config-if

Parameters

<IPV6-ADDR>

Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Enabling IPv6 link-local on the interface:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address link-local
```

ipv6 nd cache-limit

Syntax

```
ipv6 nd cache-limit <CACHELIMIT>
```

```
no ipv6 nd cache-limit [<CACHELIMIT>]
```

Description

Configures the limit on the number of neighbor entries in the ND cache.

The `no` form of this command sets the cache limit to the default value.

Command context

```
config
```

Parameters

<CACHELIMIT>

Specifies the neighbor cache entries limit. Range: 1-131072. Default: 131072.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the cache limit to 20.

```
switch(config)# ipv6 nd cache-limit 20
```

ipv6 nd dad attempts

Syntax

```
ipv6 nd dad attempts <NUM-ATTEMPTS>
```

```
no ipv6 nd dad attempts [<NUM-ATTEMPTS>]
```

Description

Configures the number of neighbor solicitations to be sent when performing duplicate address detection (DAD) for a unicast address configured on an interface.

The `no` form of this command sets the number of attempts to the default value.

Command context

```
config-if
```

Parameters

dad attempts <NUM-ATTEMPTS>

Specifies the number of neighbor solicitations to send. Range: 0-15. Default: 1.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd dad attempts 5
```

ipv6 nd hop-limit

Syntax

```
ipv6 nd hop-limit <HOPLIMIT>

no ipv6 nd hop-limit [<HOPLIMIT>]
```

Description

Configures the hop limit to be sent in RAs.

The `no` form of this command resets the hop limit to 0. This reset eliminates the hop limit from the RAs that originate on the interface, so the host determines the hop limit.

Command context

config-if

Parameters

hop-limit <HOPLIMIT>

Specifies the hop limit. Range: 0-255. Default: 64.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd hop-limit 64
```

ipv6 nd mtu

Syntax

```
ipv6 nd mtu <MTU-VALUE>

no ipv6 nd mtu [<MTU-VALUE>]
```

Description

Configures the MTU size to be sent in the RA messages.

The `no` form of this command sets hop limit to the default value.

Command context

config-if

Parameters

<MTU-VALUE>

Specifies the MTU size. Range: 1280-65535 bytes. Default: 1500 bytes.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd mtu 1300
```

ipv6 nd ns-interval

Syntax

```
ipv6 nd ns-interval <TIME>
no ipv6 nd ns-interval [<TIME>]
```

Description

Configures the ND time between DAD neighbor solicitations sent for an unresolved destination, or between duplicate address detection neighbor solicitation requests. Increase this setting when neighbor solicitation retries or failures are occurring, or in a slow (WAN) network.

The `no` form of this command sets the ns-interval to the default value.

Command context

config-if

Parameters

<TIME>

Specifies the neighbor solicitation interval. Range: 1000-3600000 milliseconds. Default: 1000 milliseconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ns-interval 1200
```

ipv6 nd prefix

Syntax

```
ipv6 nd prefix <IPV6-ADDR>/<PREFIX-LEN>
    [no-advertise | [valid <LIFETIME-VALUE> preferred
    <LIFETIME-VALUE>] | no-autoconfig | no-onlink]

no ipv6 nd prefix <IPV6-ADDR>/<PREFIX-LEN> [no-advertise
    | [valid <LIFETIME-VALUE> preferred <LIFETIME-VALUE>
    ] | no-autoconfig | no-onlink]

ipv6 nd prefix default [no-advertise | [valid <LIFETIME-VALUE>
    preferred <LIFETIME-VALUE>] | no-autoconfig | no-onlink]}
```

```
no ipv6 nd prefix default [no-advertise | [valid <LIFETIME-VALUE>
    preferred <LIFETIME-VALUE>] | no-autoconfig | no-onlink]]
```

Description

Specifies prefixes for the routing switch to include in RAs transmitted on the interface. IPv6 hosts use the prefixes in RAs to autoconfigure themselves with global unicast addresses. The autoconfigured address of a host is composed of the advertised prefix and the interface identifier in the current link-local address of the host.

By default, advertise, autoconfig, and onlink are set.

The **no** form of this command removes the configuration on the interface.

Command context

config-if

Parameters

<IPV6-ADDR>/<PREFIX-LEN>

Specifies the IPv6 prefix to advertise in RA. Format: X:X::X:X/M

default

Specifies apply configuration to all on-link prefixes that are not individually set by the **ipv6 ra prefix <IPV6-ADDR>/<PREFIX-LEN>** command. It applies the same valid and preferred lifetimes, link state, autoconfiguration state, and advertise options to the advertisements sent for all on-link prefixes that are not individually configured with a unique lifetime. This also applies to the prefixes for any global unicast addresses configured later on the same interface.

Using default once, and then using it again with any new parameter values results in the new values replacing the former values in advertisements. If default is used without the **no-advertise**, **no-autoconfig**, or **no-onlink** parameter, the advertisement setting for the absent parameter is returned to its default setting.

no-advertise

Specifies do not advertise prefix in RA.

valid <LIFETIME-VALUE>

Specifies the total time, in seconds, the prefix remains available before becoming unusable. After preferred-lifetime expiration, any autoconfigured address is deprecated and used only for transactions only before preferred-lifetime expires. If the valid lifetime expires, the address becomes invalid.

You can enter a value in seconds or enter **valid infinite** which sets infinite lifetime. Default: 2,592,000 seconds which is 30 days. Range: 0–4294967294 seconds.

preferred <LIFETIME-VALUE>

Specifies the span of time during which the address can be freely used as a source and destination for traffic. This setting must be less than or equal to the corresponding valid-lifetime setting.

You can enter a value in seconds or enter **preferred infinite** which sets infinite lifetime. Default: 604,800 seconds which is seven days. Range: 0–4294967294 seconds.

no-autoconfig

Specifies do not use prefix for autoconfiguration.

no-onlink

Specifies do not use prefix for onlink determination.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd prefix 4001::1/64 valid 30 preferred 10 no-autoconfig no-onlink
```

ipv6 nd ra dns search-list

Syntax

```
ipv6 nd ra dns search-list <DOMAIN-NAME> [lifetime <TIME>]
```

```
no ipv6 nd ra dns search-list <DOMAIN-NAME>
```

Description

Configures the DNS Search List (DNSSL) to include in Router Advertisements (RAs) transmitted on the interface.

The `no` form of this command removes the DNS Search List from the RAs transmitted on the interface.

Command context

config-if

Parameters

<DOMAIN-NAME>

Specifies the domain names for DNS queries.

lifetime <TIME>

Specifies lifetime in seconds. Range: 4-4294967295 seconds. Default: 1800 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- DNSSL contains the domain names of DNS suffixes or IPv6 hosts to append to short, unqualified domain names for DNS queries.
- Multiple DNS domain names can be added to the DNSSL by using the command repeatedly.
- A maximum of eight server addresses are allowed.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns search-list test.com lifetime 500
```

ipv6 nd ra dns server

Syntax

```
ipv6 nd ra dns server <IPV6-ADDR> [lifetime <TIME>]
```

```
no ipv6 nd ra dns server <IPV6-ADDR>
```

Description

Configures the IPv6 address of a preferred Recursive DNS Server (RDNSS) to be included in Router Advertisements (RAs) transmitted on the interface.

The `no` form of this command removes the configured DNS server from the RAs transmitted on the interface.

Command context

`config-if`

Parameters

<IPV6-ADDR>

Specifies the RDNSS address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

lifetime <TIME>

Specifies IPv6 DNS server lifetime in seconds. Range: 4-4294967295 seconds. Default: 1800 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Including RDNSS information in RAs provides DNS server configuration for connected IPv6 hosts without requiring DHCPv6.
- Multiple servers can be configured on the interface by using the command repeatedly.
- A maximum of eight server addresses are allowed.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns server 2001::1 lifetime 400
```

ipv6 nd ra lifetime

Syntax

```
ipv6 nd ra lifetime <TIME>
```

```
no ipv6 nd ra lifetime [<TIME>]
```

Description

Configures the lifetime, in seconds, for the routing switch to be used as a default router by hosts on the current interface.

The `no` form of this command sets lifetime to the default of 1800 seconds.

Command context

`config-if`

Parameters

<TIME>

Specifies lifetime in seconds of a default router. A setting of 0 for default router lifetime in an RA indicates that the routing switch is not a default router on the interface. Range: 0-9000 seconds. Default: 1800 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- A given host on an interface refreshes the default router lifetime for a specific router each time the host receives an RA from that router.
- A specific router ceases to be a default router candidate for a given host if the default router lifetime expires before the host is updated with a new RA from the router.

Examples

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ipv6 nd ra lifetime 1200
```

ipv6 nd ra managed-config-flag

Syntax

```
ipv6 nd ra managed-config-flag
```

```
no ipv6 nd ra managed-config-flag
```

Description

Controls the M flag setting in RAs the router transmits on the current interface. Enable the M flag to indicate that hosts can obtain IP address through DHCPv6. The M flag is disabled by default.

The `no` form of this command turns off (disables) the M flag.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Enabling the M flag directs hosts to acquire their IPv6 addressing for the current interface from a DHCPv6 server.
- When the M-bit is enabled, receiving hosts ignore the O flag setting, which is configured using the command `ipv6 nd ra other-config-flag`.
- When the M-bit is disabled (the default), receiving hosts expect to receive their IPv6 addresses from RA.

M flag	O flag	Description
0	0	Indicates that no information is available via DHCPv6.
0	1	Indicates that other configuration information is available via DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.
1	0	Indicates that addresses are available via Dynamic Host Configuration Protocol (DHCPv6).
1	1	If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra managed-config-flag
```

ipv6 nd ra max-interval

Syntax

```
ipv6 nd ra max-interval <TIME>
```

```
no ipv6 nd ra max-interval [<TIME>]
```

Description

Configures the maximum interval between transmissions of IPv6 RAs on the interface. The interval between RA transmissions on an interface is a random value that changes every time an RA is sent. The interval is calculated to be a value between the current max-interval and min-interval settings.

The `no` form of this command returns the setting to its default, provided the default value is less than the default lifetime value.

Command context

```
config-if
```

Parameters

<TIME>

Specifies the maximum advertisement time in seconds. Range: 4-1800. Default: 600 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- This value has one setting per interface. The setting does not apply to RAs sent in response to a router solicitation received from another device.
- Attempting to set max-interval to a value that is not sufficiently larger than the current min-interval also results in an error message.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra max-interval 30
```

ipv6 nd ra min-interval

Syntax

```
ipv6 nd ra min-interval <TIME>
```

```
no ipv6 nd ra min-interval [<TIME>]
```

Description

Configures the minimum interval between transmissions of IPv6 RAs on the interface. The interval between RA transmissions on an interface is a random value that changes every time an RA is sent. The interval is calculated to be a value between the current max-interval and min-interval settings.

The `no` form of this command returns the setting to its default, provided the default value is less than the current max-interval setting.

Command context

```
config-if
```

Parameters

<TIME>

Specifies a minimum advertisement time in seconds. Range: 3-1350. Default: 200 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- This value has one setting per interface and does not apply to RAs sent in response to a router solicitation received from another device.
- The min-interval must be less than the max-interval. Attempting to set min-interval to a higher value results in an error message.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra min-interval 25
```

ipv6 nd ra other-config-flag

Syntax

```
ipv6 nd ra other-config-flag
```

```
no ipv6 nd ra other-config-flag
```

Description

Controls the O-bit in RAs the router transmits on the current interface; but is ignored unless the M-bit is disabled in RAs. Configure to set the O-bit in RA messages for host to obtain network parameters through DHCPv6. The other-config-flag is disabled by default.

For more information on configuring the M-bit, see `ipv6 nd ra managed-config-flag`.

The `no` form of this command turns off (disables) the setting for this command in RAs.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Usage

Enabling the O-bit while the M-bit is disabled directs hosts on the interface to acquire their other configuration information from DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra other-config-flag
```

ipv6 nd ra reachable-time

Syntax

```
ipv6 nd ra reachable-time <TIME>
```

```
no ipv6 nd ra reachable-time [<TIME>]
```

Description

Sets the amount of time that the interface considers a device to be reachable after receiving a reachability confirmation from the device.

The `no` form of this command sets the reachable time to the default value of 0. (no limit).

Command context

config-if

Parameters

<TIME>

Specifies the reachable time in milliseconds. Range: 1000-3600000. Default: 0 (no limit).

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra reachable-time 2000
```

ipv6 nd ra retrans-timer

Syntax

```
ipv6 nd ra retrans-timer <TIME>
```

```
no ipv6 nd ra retrans-timer [<TIME>]
```

Description

Configures the period (retransmit timer) between ND solicitations sent by a host for an unresolved destination, or between DAD neighbor solicitation requests. By default, hosts on the interface use their own locally configured NS-interval settings instead of using the value received in the RAs.

Increase this timer when neighbor solicitation retries or failures occur, or in a "slow" (WAN) network.

The `no` form of this command sets the value to the default of 0.

Command context

```
config-if
```

Parameters

<TIME>

Specifies the retransmit timer value in milliseconds. Range: 0 - 4294967295 milliseconds. Default: 0 (Use locally configured NS-interval).

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra retrans-timer 400
```

ipv6 nd router-preference

Syntax

```
ipv6 nd router-preference {high | medium | low}
```

```
no ipv6 nd router-preference [high | medium | low]
```

Description

Specifies the value that is set in the Default Router Preference (DRP) field of Router Advertisements (RAs) that the switch sends from an interface. An interface with a DRP value of high will be preferred by other devices on the network over interfaces with an RA value of medium or low.

The `no` form of this command set the value to the default of medium.

Command context

```
config-if
```

Parameters

high

Sets DRP to high.

medium

Sets DRP to medium. Default.

low

Sets DRP to low.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd router-preference high
```

ipv6 nd suppress-ra

Syntax

```
ipv6 nd suppress-ra [<SUPPRESS-OPTION>]

no ipv6 nd ra supress-ra [<SUPPRESS-OPTION>]
```

Description

Configures suppression of IPv6 Router Advertisement transmissions on an interface.

The `no` form of this command restores transmission of IPv6 Router Advertisement and options.

Command context

config-if

Parameters

suppress-ra [<SUPPRESS-OPTION>]

Specifies suppressing RA transmissions. Entering `suppress-ra` without any options, suppresses all RA messages (default). Or you can enter one of the following options.

dnssl

Specifies suppressing DNSSL options in RA messages.

mtu

Specifies suppressing MTU options in RA messages.

rdnss

Specifies suppressing RDNSS options in RA messages.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd suppress-ra mtu dnssl rdnss
switch(config-if-vlan)# no ipv6 nd suppress-ra mtu dnssl rdnss
```

show ipv6 nd global traffic

Syntax

```
show ipv6 nd global traffic [vsx-peer]
```

Description

Displays IPV6 Neighbor Discovery traffic details on a device.

Command context

Operator (>) or Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch# show ipv6 nd global traffic
ICMPv6 packet Statistics (sent/received)
  Total Messages           :      18/0
  Error Messages           :       0/0
  Destination Unreachables :       0/0
  Time Exceeded            :       0/0
  Parameter Problems       :       0/0
  Echo Request             :       0/0
  Echo Replies             :       0/0
  Redirects                :       0/0
  Packet Too Big           :       0/0
  Router Advertisements    :       4/0
  Router Solicitations     :       0/0
  Neighbor Advertisements  :       0/0
  Neighbor Solicitations   :       3/0
  Duplicate router RA received :    0/0
ICMPv6 MLD Statistics (sent/received)
  V1 Queries :      0/0
  V2 Queries :      0/0
  V1 Reports  :      0/0
  V2 Reports  :     11/0
  V1 Leaves  :      0/0
```

show ipv6 nd interface

Syntax

```
show ipv6 nd interface [<IF-NAME> | all-vrfs | vrf <VRF-NAME>]  
[vsx-peer]
```

Description

Displays neighbor discovery information for an interface. If no options are specified, displays information for the default VRF.

Command context

Operator (>) or Manager (#)

Parameters

<IF-NAME>

Displays information about the specified IPv6 enabled interface.

all-vrfs

Displays information about interfaces in all VRFs.

vrf <VRF-NAME>

Displays information about interfaces in a particular VRF. Or, if <VRF-NAME> is not specified, information for the default VRF is displayed.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all VRFs:

```
switch# show ipv6 nd interface all-vrfs
```

```
List of IPv6 Interfaces for VRF default  
Interface vlan2 is up  
Admin state is up  
IPv6 address:  
IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]  
ICMPv6 active timers:  
  Last Router-Advertisement sent:  
  Next Router-Advertisement sent in:  
Router-Advertisement parameters:  
  Periodic interval: 200 to 600 secs  
  Router Preference: medium  
  Send "Managed Address Configuration" flag: false  
  Send "Other Stateful Configuration" flag: false  
  Send "Current Hop Limit" field: 64  
  Send "MTU" option value: 1500  
  Send "Router Lifetime" field: 1800  
  Send "Reachable Time" field: 0
```

```
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
ICMPv6 error message parameters:
    Send redirects: false
ICMPv6 DAD parameters:
    Current DAD attempt: 1
```

List of IPv6 Interfaces for VRF red

Interface vlan3 is up

Admin state is up

IPv6 address:

2001::1/64 [VALID]

IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]

ICMPv6 active timers:

Last Router-Advertisement sent:

Next Router-Advertisement sent in:

Router-Advertisement parameters:

Periodic interval: 200 to 600 secs

Router Preference: medium

Send "Managed Address Configuration" flag: false

Send "Other Stateful Configuration" flag: false

Send "Current Hop Limit" field: 64

Send "MTU" option value: 1500

Send "Router Lifetime" field: 1800

Send "Reachable Time" field: 0

Send "Retrans Timer" field: 0

Suppress RA: true

Suppress MTU in RA: true

ICMPv6 error message parameters:

Send redirects: false

ICMPv6 DAD parameters:

Current DAD attempt: 1

Showing information for interface vlan 2:

```
switch# show ipv6 nd interface vlan 2
```

Interface vlan2 is up

Admin state is up

IPv6 address:

IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]

ICMPv6 active timers:

Last Router-Advertisement sent:

Next Router-Advertisement sent in:

Router-Advertisement parameters:

Periodic interval: 200 to 600 secs

Router Preference: high

Send "Managed Address Configuration" flag: false

Send "Other Stateful Configuration" flag: false

Send "Current Hop Limit" field: 64

Send "MTU" option value: 1500

Send "Router Lifetime" field: 1800

Send "Reachable Time" field: 0

Send "Retrans Timer" field: 0

Suppress RA: true

Suppress MTU in RA: true

ICMPv6 error message parameters:

Send redirects: false

ICMPv6 DAD parameters:

Current DAD attempt: 1

Showing information for the default VRF:

```
switch# show ipv6 nd interface
```

```
List of IPv6 Interfaces for VRF default
Interface vlan2 is up
  Admin state is up
  IPv6 address:
    2001::1/64 [VALID]
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent: 6 Secs
    Next Router-Advertisement sent in: 7 Secs
  Router-Advertisement parameters:
    Periodic interval: 3 to 13 secs
    Router Preference: medium
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1900
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
  ICMPv6 error message parameters:
    Send redirects: false
  ICMPv6 DAD parameters:
    Current DAD attempt: 1
```

show ipv6 nd interface prefix

Syntax

```
show ipv6 nd interface prefix [all-vrfs | vrf <VRF-NAME>] [vsx-peer]
```

Description

Shows IPv6 prefix information for all VRFs or a specific VRF. If no options are specified, shows information for the default VRF.

Command context

Operator (>) or Manager (#)

Parameters

all-vrfs

Shows prefix information for all VRFs.

vrf <VRF-NAME>

Name of a VRF.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing prefix information for the default VRF:

```
switch# show ipv6 nd interface prefix

List of IPv6 Interfaces for VRF default
List of IPv6 Prefix advertised on vlan2
  Prefix : 4545::/65
  Enabled : Yes
  Validlife time : 2592000
  Preferred lifetime : 604800
  On-link : Yes
  Autonomous : Yes
```

Showing information for VRF red:

```
switch# show ipv6 nd interface prefix vrf red

List of IPv6 Interfaces for VRF red
List of IPv6 Prefix advertised on vlan3
  Prefix : 2001::/64
  Enabled : Yes
  Validlife time : 2592000
  Preferred lifetime : 604800
  On-link : Yes
  Autonomous : Yes
```

show ipv6 nd ra dns search-list

Syntax

```
show ipv6 nd ra dns search-list [vsx-peer]
```

Description

Displays domain name information on all interfaces.

Command context

Operator (>) or Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns search-list test.com
switch# show ipv6 nd ra dns search-list
Recursive DNS Search List on: 1
```

```
Suppress DNS Search List: Yes
DNS Search 1: test.com      lifetime 1800
```

show ipv6 nd ra dns server

Syntax

```
show ipv6 nd ra dns server [vsx-peer]
```

Description

Displays DNS server information on all interfaces.

Command context

Operator (>) or Manager (#)

Parameters

[**vsx-peer**]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns server 2001::1
switch# show ipv6 nd ra dns server
Recursive DNS Server List on: 1
    Suppress DNS Server List: Yes
    DNS Server 1: 2001::1      lifetime 1800
```

sFlow is a technology for monitoring traffic in switched or routed networks. The sFlow monitoring system is comprised of:

- An sFlow Agent that runs on a network device, such as a switch. The agent uses sampling techniques to capture information about the data traffic flowing through the device and forwards this information to an sFlow collector.
- An sFlow Collector that receives monitoring information from sFlow agents. The collector stores this information so that a network administrator can analyze it to understand network data flow patterns. One sFlow collector can receive the data from many sFlow agents.



NOTE: The sFlow UDP datagrams sent to a collector are not encrypted, therefore any sensitive information contained in an sFlow sample is exposed.

sFlow agent

The sFlow agent on the switch provides ingress sampling of all forwarded layer 2 and layer 3 traffic on LAG and Ethernet ports. High-availability is supported (packet sampling continues to work after switch-over).

The sFlow agent can communicate with up to three sFlow collectors at the same time. The agent communicates with collectors only on the default VRF.

Although you can configure very high sampling rates, the switch may drop samples if it cannot handle the rate of sampled packets. High sampling rates may also cause high CPU usage resulting in control plane performance issues.

A single sFlow datagram sent to a collector contains multiple flow and counter samples. The total number of samples an sFlow datagram can contain varies depending on the settings for header size and maximum datagram size.

Default settings

- sFlow is disabled on all interfaces.
- Collector port: UDP port 6343.
- Sampling rate: 4096.
- Polling interval: 30 seconds.
- Header size: 128 bytes.
- Max datagram size: 1400 bytes.

Supported features

- Global sampling rate
- Interface counters polling
- Agent IP configuration for IPv4 and IPv6
- Header size configuration

- Max datagram size configuration
- Ingress sampling for all forwarded traffic (L2, L3)
- Enable/Disable sFlow per interface
- Support for three remote collectors
- An out-of-band collector can be defined on the management VRF
- A collector can be defined on the non-default VRF
- Sampling on Ethernet and LAG interfaces
- High availability support (sampling continues to work after switch-over)
- Source IP support (setting source IP for sFlow datagrams sent to a remote collector)

Limitations

- No sampling of egress traffic
- Sampling rate cannot be set per interface (global only)
- sFlow is not configurable via SNMP

Configuring the sFlow agent

Procedure

1. Configure one or more sFlow collectors with the command **`sflow collector`**. This determines where the sFlow agent sends sFlow information.
2. Enable the sFlow agent on all interfaces, or on a specific interface, with the command **`sflow`**.
3. Define the address of the sFlow agent with the command **`sflow agent-ip`**.
4. By default, the source IP address for sFlow datagrams is set to the IP address of the outgoing switch interface on which the sFlow client is communicating with a collector. Since the switch can have multiple routing interfaces, datagrams can potentially be sent on different paths at different times, resulting in different source IP addresses for the same client. To resolve this issue, define a single source IP address. For details, see *Single source IP address* in the *Fundamentals Guide*.
5. For most deployments, the default values for the following settings do not need to be changed. If your deployment requires different settings, change the default values with the indicated commands:

sFlow setting	Default value	Command to change it
Rate at which packets are sampled.	1 in every 4096 packets	<code>sflow sampling</code>
Rate at which the switch sends data to an sFlow collector.	30 seconds	<code>sflow polling</code>
Size of the sFlow header.	128 bytes	<code>sflow header-size</code>
Maximum size of an sFlow datagram.	1400 bytes	<code>sflow max-datagram-size</code>

6. Review sFlow configuration settings with the command **`show sflow`**.

Example

This example creates the following configuration:

- Configures an sFlow collector with the IP address 10.10.20.209.
- Enables the sFlow agent on all interfaces.
- Defines the sFlow agent IP address to be 10.10.1.5.

```
switch(config)# sflow collector 10.10.20.209
switch(config)# sflow
switch(config)# sflow agent-ip 10.0.0.1
```

sFlow scenario

In this scenario, two hosts send sFlow traffic through a switch to an sFlow collector.

Procedure

1. Enable sFlow globally.

```
switch# config
switch(config)# sflow
```

2. Set the sFlow agent IP address to 10.10.12.1.

```
switch(config)# sflow agent-ip 10.10.12.1
```

3. Set the sFlow collector IP address to 10.10.12.2.

```
switch(config)# sflow collector 18.2.2.2
```

4. Configure sFlow sampling rate and polling interval.

```
switch(config)# sflow sampling 5000
switch(config)# sflow polling 20
```

5. Configure interface **vlan 2** with IP address 10.10.10.1/24.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.10.1/24
switch(config)# quit
```

6. Configure interface **vlan 3** with IP address 10.10.11.1/24.

```
switch(config)# interface vlan 3
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.11.1/24
switch(config)# quit
```

7. Configure interface **vlan 4** with IP address 10.10.12.1/24.

```
switch(config)# interface vlan 4
switch(config-if-vlan)# no shutdown
```

```
switch(config-if-vlan)# ip address 10.10.12.1/24
switch(config)# quit
```

8. Verify sFlow configuration.

```
switch# show sflow

sFlow Global Configuration
-----
sFlow                      enabled
Collector IP/Port/Vrf      10.10.12.2/6343/default
Agent Address              10.10.12.1
Sampling Rate              5000
Polling Interval           20
Header Size                128
Max Datagram Size          1400

sFlow Status
-----
Running - Yes

sFlow Statistics
-----
Number of Samples          25
```

sFlow scenario 2

In this scenario, two hosts connected to different switches send sFlow traffic to a collector. A LAG is used to connect the two switches.

Procedure

1. Configure switch 1.

a. Enable sFlow globally.

```
switch# config
switch(config)# sflow
```

b. Set the sFlow agent IP address to 10.10.12.1.

```
switch(config)# sflow agent-ip 10.10.12.1
```

c. Set the sFlow collector IP address to 10.10.12.2.

```
switch(config)# sflow collector 10.10.12.2
```

d. Configure sFlow sampling rate and polling interval.

```
switch(config)# sflow sampling 5000
switch(config)# sflow polling 10
```

e. Create VLAN 8.

```
switch(config)# vlan 8
switch(config-vlan-8)# no shutdown
switch(config)# exit
```

- f. Define LAG 100 and assign VLAN `vlan 8` to it.

```
switch(config)# interface lag 100
switch(config-lag-if)# no shutdown
switch(config-lag-if)# vlan access 8
switch(config-lag-if)# lacp mode active
```

- g. Configure interface `1/1/1`.

```
switch(config)# interface 1/1/1
switch(config-if-vlan)# no shutdown
switch(config-lag-if)# no routing
switch(config-if-vlan)# vlan access 8
```

- h. Configure interface `1/1/2` and `1/1/3` as members of LAG 100.

```
switch# (config)#interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# lag 100
switch(config-if)# exit
switch(config-if-vlan)# interface 1/1/3
switch(config-if)# no shutdown
switch(config-if)# lag 100
switch(config-if)# exit
```

- i. Configure interface `vlan 5` with IP address `10.10.12.1/24`.

```
switch# (config)#interface vlan 5
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.12.1/24
switch(config-if-vlan)# quit
```

- j. Verify sFlow configuration.

```
switch# show sflow
```

```
sFlow Global Configuration
```

```
-----
sFlow                               enabled
Collector IP/Port/Vrf               10.10.12.2/6343/default
Agent Address                       10.10.12.1
Sampling Rate                       5000
Polling Interval                    10
Header Size                         128
Max Datagram Size                   1400
```

```
sFlow Status
```

```
-----
Running - Yes
```

```
sFlow Statistics
```

```
-----
Number of Samples                   120
```

2. Configure switch 2.

a. Create VLAN 8.

```
switch(config)# vlan 8  
switch(config-vlan-8)# no shutdown  
switch(config)# exit
```

b. Define LAG 100 and assign VLAN `vlan 8` to it.

```
switch(config)# interface lag 100  
switch(config-lag-if)# no shutdown  
switch(config-lag-if)# vlan access 8  
switch(config-lag-if)# lacp mode active
```

c. Configure interface 1/1/1.

```
switch(config)# interface 1/1/1  
switch(config-if)# no shutdown  
switch(config-if)# vlan access 8
```

d. Configure interface 1/1/2 and 1/1/3 as members of LAG 100.

```
switch# (config)#interface 1/1/2  
switch(config-if)# no shutdown  
switch(config-if)# lag 100  
switch(config-if)# exit  
switch(config)-if# interface 1/1/3  
switch(config-if)# no shutdown  
switch(config-if)# lag 100  
switch(config-if)# exit
```

sFlow agent commands

sflow

Syntax

```
sflow
```

```
no sflow
```

Description

Enables the sFlow agent.

- In the `config` context, this command enables the sFlow agent globally on all interfaces.
- In an `config-if` context, this command enables the sFlow agent on a specific interface. sFlow cannot be enabled on a member of a LAG, only on the LAG.

The sFlow agent is disabled by default.

The `no` form of this command disables the sFlow agent and deletes all sFlow configuration settings, either globally, or for a specific interface.

Command context

```
config
```

config-if

Parameters

None.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling sFlow globally on all interfaces:

```
switch(config)# sflow
```

Disabling sFlow globally on all interfaces:

```
switch(config)# no sflow
```

Enabling sFlow on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# sflow
```

Disabling sFlow on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no sflow
```

Enabling sFlow on interface lag100:

```
switch(config)# interface lag100
switch(config-if)# sflow
```

Disabling sFlow on interface lag100:

```
switch(config)# interface lag100
switch(config-if)# no sflow
```

sflow agent-ip

Syntax

```
sflow agent-ip <IP-ADDR>
```

```
no sflow agent-ip [<IP-ADDR>]
```

Description

Defines the IP address of the sFlow agent to use in sFlow datagrams. This address must be defined for sFlow to function. HPE recommends that the address:

- can uniquely identify the switch
- is reachable by the sFlow collector
- does not change with time

The `no` form of this command deletes the IP address of the sFlow agent. This causes sFlow to stop working and no datagrams will be sent to the sFlow collector.

Command context

config

Parameters

<IP-ADDR>

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. The agent address is used to identify the switch in all sFlow datagrams sent to sFlow collectors. It is usually set to an IP address on the switch that is reachable from an sFlow collector.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the agent address to 10.10.10.100:

```
switch(config)# sflow agent-ip 10.0.0.100
```

Setting the agent address to 2001:0db8:85a3:0000:0000:8a2e:0370:7334:

```
switch(config)# sflow agent-ip 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Removing the address configuration from the switch, which results in sFlow being disabled:

```
switch(config)# no sflow agent-ip
```

sflow collector

Syntax

```
sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
```

```
no sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
```

Description

Defines a collector to which the sFlow agent sends data. Up to three collectors can be defined. At least one collector should be defined, and it must be reachable from the switch for sFlow to work.

Command context

config

Parameters

collector <IP-ADDR>

Specifies the IP address of a collector in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

port <PORT>

Specifies the UDP port on which to send information to the sFlow collector. Range: 0 to 65536. Default: 6343.

vrf <VRF>

Specifies the VRF on which to send information to the sFlow collector. The VRF must be defined on the switch. If no VRF is specified, the default VRF (default) is used.

Authority

Administrators or local user group members with execution rights for this command.

Example

Defining a collector with IP address 10.10.10.100 on UDP port 6400:

```
switch(config)# sflow collector 10.0.0.1 port 6400
```

sflow disable

Syntax

```
sflow disable
```

Description

Disables the sFlow agent, but retains any existing sFlow configuration settings. The settings become active if the sFlow agent is re-enabled.

Command context

```
config
```

Parameters

None.

Authority

Administrators or local user group members with execution rights for this command.

Example

Disabling sFlow support:

```
switch(config)# sflow disable
```

sflow header-size

Syntax

```
sflow header-size <SIZE>
```

```
no sflow header-size [<SIZE>]
```

Description

Sets the sFlow header size in bytes.

The `no` form of this command sets the header size to the default value of 128.

Command context

```
config
```

Parameters

header-size <SIZE>

Specifies the sFlow header size in bytes. Range: 64 to 256. Default: 128.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the header size to 64 bytes:

```
switch(config)# sflow header-size 64
```

Setting the header size to the default value of 128 bytes:

```
switch(config)# no sflow header-size
```

sflow max-datagram-size

Syntax

```
sflow max-datagram-size <SIZE>
```

```
no sflow max-datagram-size [<SIZE>]
```

Description

Sets the maximum number of bytes that are sent in one sFlow datagram.

The `no` form of this command sets maximum number of bytes to the default value of 1400.

Command context

```
config
```

Parameters

max-datagram-size <SIZE>

Specifies the maximum datagram size in bytes. Range: 1 to 9000. Default: 1400.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the datagram size to 1000 bytes:

```
switch(config)# sflow max-datagram-size 1000
```

Setting the header size to the default value of 1400 bytes:

```
switch(config)# no sflow max-datagram-size
```

sflow polling

Syntax

```
sflow polling <INTERVAL>
```

```
no sflow polling [<INTERVAL>]
```

Description

Defines the global polling interval for sFlow in seconds.

The `no` form of this command sets the polling interval to the default value of 30 seconds.

Command context

`config`

Parameters

<INTERVAL>

Specifies the polling interval in seconds. Range: 10 to 3600. Default: 30.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the polling interval to 10:

```
switch(config)# sflow polling 10
```

Setting the polling interval to the default value.

```
switch(config)# no sflow polling
```

sflow sampling

Syntax

```
sflow sampling <RATE>
```

```
no sflow sampling [<RATE>]
```

Description

Defines the global sampling rate for sFlow in number of packets. The default sampling rate is 4096, which means that one in every 4096 packets is sampled. A warning message is displayed when the sampling rate is set to less than 4096 and proceeds only after user confirmation.

The `no` form of this command sets the sampling rate to the default value of 4096.

Command context

`config`

Parameters

sampling <RATE>

Specifies the sampling rate. Range: 1 to 1000000000. Default: 4096.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the sampling rate to 5000:

```
switch(config)# sflow sampling 5000
```

Setting the sampling rate to the default:

```
switch(config)# no sflow sampling
```

Setting the sampling rate to 1000:

```
switch(config)# sflow sampling 1000
Setting the sFlow sampling rate lower than 4096 is not recommended and might
affect system performance.
Do you want to continue [y/n]? y
switch(config)#
```

show sflow

Syntax

```
show sflow [interface <INTERFACE-NAME>] [vsx-peer]
```

Description

Shows sFlow configuration settings and statistics for all interfaces, or for a specific interface

Command context

Manager (#)

Parameters

interface <INTERFACE-NAME>

Specifies the name of an interface on the switch.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing sFlow information for all interfaces:

```
switch# show sflow
sFlow Global Configuration
-----
sFlow                      enabled
Collector IP/Port/Vrf      10.0.0.2/6343/default
                           10.0.0.3/6400/default
Agent Address              10.0.0.1
Sampling Rate              1024
Polling Interval           30
Header Size                128
Max Datagram Size          1400

sFlow Status
-----
Running - Yes

sFlow Statistics
-----
Number of Samples          200
```

- Agent address is not configured.

Showing sFlow information for interface **1/1/1**:

```
switch# show sflow 1/1/1
```

```
sFlow configuration - Interface 1
```

```
-----  
sFlow                enabled  
Sampling Rate        1024  
Number of Samples    30
```

DHCP client commands

`ip dhcp`

Syntax

```
ip dhcp
```

Description

Enables the DHCP client on the management interface enabling the interface to automatically obtain an IP address from a DHCP server on the network. By default, the DHCP client is enabled.

Command context

```
config-if-vlan
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example enables the DHCP client on the management interface.

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# no shutdown
```

If the interface is not enabled, you can enable it by entering the `no shutdown` command.

The Internet Control Message Protocol (ICMP) is a supporting protocol in the Internet protocol suite. The protocol is used by network devices, including routers, to send error messages and operational information. For example, an ICMP message might indicate that a requested service is not available. Another example of an ICMP message might be that a host or router could not be reached.

ICMP message types

The type field identifies the type of message sent by the host or gateway.

Type	ICMP messages
0	Echo Reply (Ping Reply, used with Type 8, Ping Request)
3	Destination Unreachable
4	Source Quench
5	Redirect
8	Echo Request (Ping Request, used with Type 0, Ping Reply)
9	Router Advertisement (Used with Type 9)
10	Router Solicitation (Used with Type 10)
11	Time Exceeded
12	Parameter Problem
13	Timestamp Request (Used with Type 14)
14	Timestamp Reply (Used with Type 13)
15	Information Request (obsolete) (Used with Type 16)
16	Information Reply (obsolete) (Used with Type 15)
17	Address Mask Request (Used with Type 17)
18	Address Mask Reply (Used with Type 18)

When ICMP messages are sent

ICMP messages are sent when one or more of the following scenarios occur:

- A datagram cannot reach its destination.
- The gateway does not have the buffering capacity to forward a datagram.
- The gateway can direct the host to send traffic on a shorter route.

ICMP redirect messages

ICMP redirect messages are used by routers to notify the hosts on the data link that a better route is available for a particular destination.

When ICMP redirect messages are sent

The switch is configured to send redirects by default. ICMP redirect messages are sent when one or more of the following scenarios occur:

- The interface on which the packet comes into the router is the same interface on which the packet gets routed out.
- The subnet or network of the source IP address is on the same subnet or network of the next-hop IP address of the routed packet.
- The datagram is not source-routed.
- The destination unicast address is unreachable. In this case, the router generates the ICMP destination unreachable message to inform the source host about the situation.

ICMP commands

`ip icmp redirect`

Syntax

```
ip icmp redirect
no ip icmp redirect
```

Description

Enables the sending of ICMPv4 and ICMPv6 redirect messages to the source host. Enabled by default. The `no` form of this command disables ICMPv4 and ICMPv6 redirect messages to the source host.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling ICMP redirect messages:

```
switch(config)# ip icmp redirect
```

Disabling ICMP redirect messages:

```
switch(config)# no ip icmp redirect
```

ip icmp throttle

Syntax

```
ip icmp throttle <packet-interval>
```

```
no ip icmp throttle
```

Description

Used to configure the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

The `no` form of this command disables the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

Command context

```
config
```

Parameters

<packet-interval>

Specifies the ICMPv4/v6 packet interval in seconds. Default: 1 second. Range: 1-86400.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# ip icmp throttle 3000
```

Disabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# no ip icmp throttle
```

ip icmp unreachable

Syntax

```
ip icmp unreachable
```

```
no ip icmp unreachable
```

Description

Enables the sending of ICMPv4 and ICMPv6 destination unreachable messages on the switch to a source host when a specific host is unreachable. The unreachable host address originates from the failed packet. Default setting.

The `no` form of this command disables the sending of ICMPv4 and ICMPv6 destination unreachable messages from the switch to a source host when a specific host is unreachable. This command does not prevent other hosts from sending an ICMP unreachable message.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# ip icmp unreachable
```

Disabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# no ip icmp unreachable
```


The Domain Name System (DNS) is the Internet protocol for mapping a hostname to its IP address. DNS allows users to enter more readily memorable and intuitive hostnames, rather than IP addresses, to identify devices connected to a network. It also allows a host to keep the same hostname even if it changes its IP address.

Hostname resolution can be either static or dynamic.

- In static resolution, a local table is defined on the switch that associates hostnames with their IP addresses. Static tables can be used to speed up the resolution of frequently queried hosts.
- Dynamic resolution requires that the switch query a DNS server located elsewhere on the network. Dynamic name resolution takes more time than static name resolution, but requires far less configuration and management.

DNS client

The DNS client resolves hostnames to IP addresses for protocols that are running on the switch. When the DNS client receives a request to resolve a hostname, it can do so in one of two ways:

- Forward the request to a DNS name server for resolution.
- Reply to the request without using a DNS name server, by resolving the name using a statically defined table of hostnames and their associated IP addresses.

Configuring the DNS client

Procedure

1. Configure one or more DNS name servers with the command `ip dns server`.
2. To resolve DNS requests by appending a domain name to the requests, either configure a single domain name with the command `ip dns domain-name`, or configure a list of up to six domain names with the command `ip dns domain-list`.
3. To use static name resolution for certain hosts, associate an IP address to a host with the command `ip dns host`.
4. Review your DNS configuration settings with the command `show ip dns`.

Examples

This example creates the following configuration:

- Defines the domain **switch.com** to append to all requests.
- Defines a DNS server with IPv4 address of **1.1.1.1**.

- Defines a static DNS host named **myhost1** with an IPv4 address of **3.3.3.3**.
- DNS client traffic is sent on the default VRF (named **default**).

```
switch(config)# ip dns domain-name switch.com
switch(config)# ip dns server-address 1.1.1.1
switch(config)# ip dns host myhost1 3.3.3.3
switch(config)# exit
switch# show ip dns
```

```
VRF Name : default
Domain Name: switch.com
Name Server(s): 1.1.1.1
```

Host Name	Address
myhost1	3.3.3.3

DNS client commands

ip dns domain-list

Syntax

```
ip dns domain-list <DOMAIN-NAME> [vrf default]
```

```
no ip dns domain-list <DOMAIN-NAME> [vrf default]
```

Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The **no** form of this command removes a domain from the list.

Command context

```
config
```

Parameters

list <DOMAIN-NAME>

Specifies a domain name. Up to six domains can be added to the list. Length: 1 to 256 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```
switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com
```

This example removes the entry **domain1.com**.

```
switch(config)# no ip dns domain-list domain1.com
```

ip dns domain-name

Syntax

```
ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

```
no ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command **ip dns domain-list**, the domain name defined with this command is ignored.

The **no** form of this command removes the domain name.

Command context

config

Parameters

<DOMAIN-NAME>

Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.

vrf <VRF-NAME>

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the default domain name to domain.com:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name domain.com:

```
switch(config)# no ip dns domain-name domain.com
```

ip dns host

Syntax

```
ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
```

```
no ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a static IP address associated with a hostname.

Command context

config

Parameters

host <HOST-NAME>

Specifies the name of a host. Length: 1 to 256 characters.

<IP-ADDR>

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

vrf <VRF-NAME>

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example defines an IPv4 address of 3.3.3.3 for host1.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of b::5 for host 1.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for host 1 with address b::5.

```
switch(config)# no ip dns host host1 b::5
```

ip dns server address

Syntax

```
ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

```
no ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The no form of this command removes a name server from the list.

Command context

config

Parameters

<IP-ADDR>

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

vrf <VRF-NAME>

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example defines a name server at 1.1.1.1.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at a::1.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at a::1.

```
switch(config)# no ip dns server-address a::1
```

show ip dns

Syntax

```
show ip dns [vrf <VRF-NAME>] [vsx-peer]
```

Description

Shows all DNS client configuration settings or the settings for a specific VRF.

Command context

Manager (#)

Parameters

vrf <VRF-NAME>

Specifies the VRF for which to show information. If no VRF is defined, the default VRF is used.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

ARP (Address Resolution Protocol) is used to map the network address assigned to a device to its physical address. For example, on an Ethernet network, ARP maps layer 3 IPv4 network addresses to layer 2 MAC addresses. (ARP does not work with IPv6 addresses. Instead, the Neighbor discovery protocol is used.)

ARP operates at layer 2. ARP requests are broadcast to all devices on the local network segment and are not forwarded by routers. ARP is enabled by default and cannot be disabled.

Proxy ARP

Proxy ARP allows a routing switch to answer ARP requests from devices on one network on behalf of devices on another network. The ARP proxy is aware of the location of the traffic destination, and offers its own MAC address as the final destination.

For example, if Proxy ARP is enabled on a routing switch connected to two subnets (10.10.10.0/24 and 20.20.20.0/24), the routing switch can respond to an ARP request from 10.10.10.69 for the MAC address of the device with IP address 20.20.20.69.

Typically, the host that sent the ARP request then sends its packets to the switch that has the ARP proxy. This switch then forwards the packets to the intended host through a mechanism such as a tunnel.

Proxy ARP is supported on L3 physical and VLAN interfaces. It is disabled by default. To enable proxy ARP, routing must be enabled on the interface.

Local proxy ARP

Local proxy ARP is a technique by which a device on a given network answers the ARP queries for a host address that is on the same network. It is primarily used to enable layer 3 communication between hosts within a common subnet that are separated by layer 2 boundaries (Example: PVLAN). Local proxy ARP is supported on L3 physical and VLAN interfaces.

Local proxy ARP is disabled by default. Routing must be enabled on the interface to enable local proxy ARP.

Dynamic ARP Inspection

ARP is used for resolving IP against MAC addresses on a broadcast network segment like the Ethernet and was originally defined by Internet Standard RFC 826. ARP does not support any inherent security mechanism and as such depends on simple datagram exchanges for the resolution, with many of these being broadcast.

Because it is an unreliable and non-secure protocol, ARP is vulnerable to attacks. Some attacks may be targeted toward the networks whereas other attacks may be targeted toward the switch itself. The attacks primarily intend to create denial of service (DoS) for the other entities present in the network.

Most of the attacks are carried out in one of the following three forms:

- Overwhelming the switch control plane with too many ARP packets.
- Overwhelming the switch control plane with too many unresolved data packets.
- Masquerading as a trusted gateway/server by wrongly advertising ARPs.

Several defense mechanisms can be put in place on a switch to protect against attacks:

- Limit the amount of ARP activity allowed from a host or on a port.
- Ensure that all ARP packets are consistent with one or more binding databases, which can be created through various means.
- Enforce integrity checks on the ARP packets to check against different MAC or IP addresses in the Ethernet or IP header and ARP header.

This release implements Dynamic ARP Inspection to enforce DHCP snooping binding on all ARP packets and is limited to the 8400 platform. The feature will be disabled from the code, CLI, and schema by the use of appropriate config flags for other platforms.

Only the following is supported:

- Enabling and disabling of Dynamic ARP Inspection on a VLAN level (it does not have to be SVI).
- Defining the member ports of a VLAN as either trusted or untrusted.
- Only ARP traffic on untrusted ports subjected to checks.
- Routed ports (RoPs) always treated as trusted.
- Listening to the DHCP Bindings table and check every ARP packet to match against the binding.

ARP ACLs are not supported in this release and the DHCP snooping table will be the only source of binding.

ARP commands

arp cache-limit

Syntax

```
arp cache-limit <LIMIT>
```

Description

Specifies the maximum number of entries in the ARP (Address Resolution Protocol) cache.

Command context

```
config
```

Parameters

<LIMIT>

Specifies the maximum number of entries in the ARP cache. Range: 4096 to 131072. Default: 131072.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# arp cache-limit 4097
```

arp inspection

Syntax

```
arp inspection
```

Description

Enables Dynamic ARP Inspection on the current VLAN, forcing all ARP packets from untrusted ports to be subjected to a MAC-IP association check against a binding table.

The `no` form of this command disables Dynamic ARP Inspection on the VLAN.

Command context

```
config-vlan-<VLAN-ID>
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling dynamic ARP inspection:

```
switch# configure terminal  
switch(config)# vlan 1  
switch(config-vlan)# arp inspection
```

Disabling dynamic ARP inspection:

```
switch# configure terminal  
switch(config)# vlan 1  
switch(config-vlan)# no arp inspection
```

arp inspection trust

Syntax

```
arp inspection trust
```

```
no arp inspection trust
```

Description

Configures the interface as a trusted. All interfaces are untrusted by default.

The `no` form of this command returns the interface to the default state (untrusted).

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting an interface as trusted:

```
switch(config-if)# arp inspection trust
```


arp ipv4 mac

Syntax

```
arp ipv4 <IPV4_ADDR> mac <MAC_ADDR>

no arp ipv4 <IPV4_ADDR> mac <MAC_ADDR>
```

Description

Specifies a permanent static neighbor entry in the ARP table (for IPv4 neighbors).
The `no` form of this command deletes a permanent static neighbor entry from the ARP table.

Command context

```
config-if
config-if-vlan
```

Parameters

ipv4 <IPV4-ADDR>

Specifies the IP address of the neighbor or the virtual IP address of the cluster in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. . Range: 4096 to 131072. Default: 131072.

mac <MAC-ADDR>

Specifies the MAC address of the neighbor or the multicast MAC address in IANA format (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring a static ARP entry on a interface VLAN 10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# arp ipv4 2.2.2.2 mac 01:00:5e:00:00:01
```

Removing a static ARP entry on interface VLAN10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no arp ipv4 2.2.2.2 mac 01:00:5e:00:00:01
```

clear arp

Syntax

```
clear arp [port <PORT-ID> | vrf {all-vrfs | <VRF-NAME>}]
```

Description

Clears IPv4 and IPv6 neighbor entries from the ARP table. If you do not specify any parameters, ARP table entries are cleared for the default VRF.

Command context

Manager (#)

Parameters

port <PORT-ID>

Specifies a physical port on the switch. Format: member/slot/port. For example: `vlan 2.`

all-vrfs

Selects all VRFs.

<VRF-NAME>

Specifies the name of a VRF. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing all IPv4 and IPv6 neighbor ARP entries for the default VRF:

```
switch# clear arp
```

Clearing all ARP neighbor entries for a port ():

```
switch# clear arp vlan 2
```

ipv6 neighbor mac

Syntax

```
ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
```

```
no ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
```

Description

Specifies a permanent static neighbor entry in the ARP table (for IPv6 neighbors).

The `no` form of this command deletes a permanent static neighbor entry from the ARP table.

Command context

config-if

Parameters

<IPV6-ADDR>

Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

mac <MAC-ADDR>

Specifies the MAC address of the neighbor (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Authority

Administrators or local user group members with execution rights for this command.

Example

Creates a static ARP entry on interface `vlan 2`.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# arp ipv6 neighbor 2001:0db8:85a3::8a2e:0370:7334 mac 00:50:56:96:df:c8
```

show arp

Syntax

```
show arp [vsx-peer]
```

Description

Shows the entries in the ARP (Address Resolution Protocol) table.

Command context

Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command displays information about ARP entries, including the IP address, MAC address, port, and state.

When no parameters are specified, the `show arp` command shows all ARP entries for the default VRF (Virtual Router Forwarding) instance.

Examples

```
switch# show arp
```

IPv4 Address Port	MAC	State	Port	Physical	
192.168.1.2	00:50:56:96:7b:e0		vlan10	1/1/29	stale
192.168.1.3	00:50:56:96:7b:ac		vlan10	1/1/1	reachable

Total Number Of ARP Entries Listed- 2.

show arp inspection interface

Syntax

```
show arp inspection interface
```

Description

Displays the current configuration of dynamic ARP inspection on a VLAN or interface.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch# show arp inspection interface
```

Interface	Trust-State
1/1/1	Untrusted

```
switch# show arp inspection interface vsx-peer
```

Interface	Trust-State
1/1/1	Untrusted
lag100	Trusted

```
switch# show arp inspection interface 1/1/1
```

Interface	Trust-State
1/1/1	Untrusted

show arp inspection statistics

Syntax

```
show arp inspection statistics
```

Description

Displays statistics about forwarded and dropped ARP packets.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch# show arp inspection statistics vlan 1-200
```

VLAN	Name	Forwarded	Dropped
1	DEFAULT_VLAN_1	0	0

```
switch# show arp inspection statistics vlan vsx-peer
```

VLAN	Name	Forwarded	Dropped
1	DEFAULT_VLAN_1	0	0
200	VLAN200	0	0

show arp state

Syntax

```
show arp state {all | failed | incomplete | permanent | reachable | stale} [vsx-peer]
```

Description

Shows ARP (Address Resolution Protocol) cache entries that are in the specified state.

Command context

Operator (>) or Manager (#)

Parameters

all

Shows the ARP cache entries for all VRF (Virtual Router Forwarding) instances.

failed

Shows the ARP cache entries that are in `failed` state. The neighbor might have been deleted.

incomplete

Shows the ARP cache entries that are in `incomplete` state.

An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. A solicitation request was sent, and the switch is waiting for a solicitation reply or a timeout.

permanent

Shows the ARP cache entries that are in `permanent` state. ARP entries that are in a permanent state can be removed by administrative action only.

reachable

Shows the ARP cache entries that are in `reachable` state, meaning that the neighbor is known to have been reachable recently.

stale

Shows ARP cache entries that are in `stale` state.

ARP cache entries are in the `stale` state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch# show arp state failed
```

IPv4 Address	MAC	Port	Physical Port	State
192.168.1.4		vlan10		failed

show arp summary

Syntax

```
show arp summary [all-vrfs | vrf <VRF-NAME>] [vsx-peer]
```

Description

Shows a summary of the IPv4 and IPv6 neighbor entries on the switch for all VRFs or a specific VRF.

Command context

Operator (>) or Manager (#)

Parameters

all-vrfs

Selects all VRFs.

vrf <VRF-NAME>

Specifies the name of a VRF.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing summary ARP information for all VRFs:

```
switch# show arp summary all-vrfs
```

ARP Entry's State	: IPv4	IPv6
Number of Reachable ARP entries	: 2	0
Number of Stale ARP entries	: 0	0
Number of Failed ARP entries	: 2	2

```

Number of Incomplete ARP entries : 0          0
Number of Permanent ARP entries  : 0          0
-----
Total ARP Entries: 6              : 4          2
-----

```

Showing a summary of all IPv4 and IPv6 neighbor entries on the primary and secondary (peer) switches:

```

vsx-primary# show arp summary
ARP Entry's State      IPv4      IPv6
-----
Number of Reachable ARP entries  25858     32231
Number of Stale ARP entries      0         1
Number of Failed ARP entries     0        257
Number of Incomplete ARP entries 0         0
Number of Permanent ARP entries  0         0
-----
Total ARP Entries- 58347         25858     32489

vsx-primary# show arp summary vsx-peer
ARP Entry's State      IPv4      IPv6
-----
Number of Reachable ARP entries  25858     32168
Number of Stale ARP entries      0         3
Number of Failed ARP entries     0        317
Number of Incomplete ARP entries 0         0
Number of Permanent ARP entries  0         0
-----
Total ARP Entries- 58346         25858     32488
-----

```

show arp timeout

Syntax

```
show arp timeout [<INTERFACE>] [vsx-peer]
```

Description

Shows the age-out period for each ARP (Address Resolution Protocol) entry for a port, LAG, or VLAN interface.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE>

Specifies a physical port, VLAN, or LAG on the switch. For physical ports, use the format (for example, 1/3/1).

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing ARP timeout information for a VLAN:

```
switch# show arp timeout vlan2
```

Port	VRF	Timeout

vlan2	default	1800

show arp vrf

Syntax

```
show arp {all-vrfs | vrf <VRF-NAME>} [vsx-peer]
```

Description

Shows the ARP table for all VRF instances, or for the named VRF.

Command context

Operator (>) or Manager (#)

Parameters

all-vrfs

Specifies all VRFs.

vrf <VRF-NAME>

Specifies the name of a VRF. Length: 1 to 32 alphanumeric characters.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing ARP entries for VRF test.

```
switch# show arp vrf test
```

ARP IPv4 Entries:

IPv4 Address	MAC	Port	Physical Port	State	VRF
10.20.30.40	00:50:56:bd:6a:c5	1/1/29	1/1/29	reachable	test

Total Number Of ARP Entries Listed: 1.

```
switch# show arp all-vrfs
```

ARP IPv4 Entries:


```

-----
IPv4 Address      MAC                Port    Physical Port  State    VRF
192.168.120.10    00:50:56:bd:10:be 1/1/32  1/1/32         reachable red
10.20.30.40       00:50:56:bd:6a:c5 1/1/29  1/1/29         reachable test
-----
Total Number Of ARP Entries Listed: 2.
-----

```

Showing ARP entries for all VRFs.

```

switch# show arp all-vrfs
ARP IPv4 Entries:
-----
IPv4 Address      MAC                Port    Physical Port  State    VRF
192.168.120.10    00:50:56:bd:10:be 1/1/32  1/1/32         reachable red
10.20.30.40       00:50:56:bd:6a:c5 1/1/29  1/1/29         reachable test
-----
Total Number Of ARP Entries Listed: 2.
-----

```

show ipv6 neighbors

Syntax

```
show ipv6 neighbors {all-vrfs | vrf <VRF-NAME>} [vsx-peer]
```

Description

Shows entries in the ARP table for all IPv6 neighbors for all VRFs or for a specific VRF.

When no parameters are specified, this command shows all ARP entries for the default VRF, and state information for `reachable` and `stale` entries only.

Command context

Operator (>) or Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Parameters

all-vrfs

Specifies all VRFs.

vrf <VRF-NAME>

Specifies a VRF name. Length: 1 to 32 alphanumeric characters.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

```

switch# show ipv6 neighbors
IPv6 Entries:

```

```

-----
IPv6 Address      MAC                Port    Physical Port  State

```

```

fe80::a21d:48ff:fe8f:2700    a0:1d:48:8f:27:00    vlan2300    1/1/31    reachable
fe80::f603:43ff:fe80:a600    f4:03:43:80:a6:00    vlan2300    1/1/30    reachable
-----

Total Number Of IPv6 Neighbors Entries Listed: 2.
-----

```

show ipv6 neighbors state

Syntax

```
show ipv6 neighbors state {all | failed | incomplete | permanent | reachable | stale} [vsx-peer]
```

Description

Shows all IPv6 neighbor ARP (Address Resolution Protocol) cache entries, or those cache entries that are in the specified state.

Command context

Operator (>) or Manager (#)

Parameters

all

Shows all ARP cache entries.

failed

Shows ARP cache entries that are in `failed` state. The neighbor might have been deleted. Set the neighbor to be unreachable.

incomplete

Shows ARP cache entries that are in `incomplete` state.

An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. This means that a solicitation request was sent, and you are waiting for a solicitation reply or a timeout.

permanent

Shows ARP cache entries that are in `permanent` state.

reachable

Shows ARP cache entries that are in `reachable` state, meaning that the neighbor is known to have been reachable recently.

stale

Shows ARP cache entries that are in `stale` state.

ARP cache entries are in the `stale` state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 neighbors state all
```

IPv6 Address	MAC	Port	Physical Port	State
100::2	48:0f:cf:af:f1:cc	lag1	lag1	reachable
300::3	48:0f:cf:af:33:be	vlan3	1/4/20	reachable
fe80::4a0f:cfff:feaf:f1cc	48:0f:cf:af:f1:cc	lag1	lag1	reachable
200::3	48:0f:cf:af:33:be	1/4/11	1/4/11	reachable
fe80::4a0f:cfff:feaf:33be	48:0f:cf:af:33:be	vlan3	1/4/20	reachable

Total Number Of IPv6 Neighbors Entries Listed- 5.

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing updates

To download product updates:

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>



IMPORTANT: Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

Documentation feedback

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